

CONTACT

Schmalkalden, Germany
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SKILLS

- Codesys, Beckhoff and TIA Portal.
- CODESYS SoftMotion & Depictor
- Github
- Robot–PLC Integration
- Python, C++
- Fusion 360, SolidWorks
- IEC 61131-3 Structured Text
- TCP/IP Socket Programming
- Camera-to-Robot Calibration
- English (C1), Deutsch (B1)

EDUCATION

Mechatronics & Robotics (M.Eng.)
Hochschule Schmalkalden, Germany
Oct 2023 - Oct 2026

- Specialization in Industrial Automation Engineering.
- Thesis on "Development and Implementation of Robotic Workcell with PLC, Digital Twin and Machine Vision for Automated material handling."

Mechanical Engineering (B.Eng.)
Visvesvaraya University (VTU), India
July 2018 - July 2022

- Specialization in Mechatronics.
- Thesis on “Terrestrial Robots for extreme environmental conditions”.

SRINIDHI SHANKAR JAIN

WORK EXPERIENCE

Master Thesis Student Feb 2026 - Sep 2026
Pilz GmbH & Co. KG and Hochschule Schmalkalden
Schmalkalden, Germany
Video Link - Pilz Robot Master’s Thesis

- Developed and tested a robotic pick-and-place system using CODESYS, CODESYS SoftMotion, and IEC 61131-3 Structured Text, including Cartesian motion sequences, conveyor handling, and automated pick/place logic.
- Built a machine-vision pipeline using Intel RealSense, Python, OpenCV, and VS Code for ROI-based workpiece detection, contour processing, localization, and orientation estimation.
- Implemented camera-to-robot calibration, converting image-space (u,v) coordinates into robot-space (X,Y,Z) coordinates and generating pose data for accurate robotic placement.
- Established raw TCP/IP socket communication between Python and CODESYS for real-time exchange of object-detection events, pose data, ROI status, and robot-control handshakes.
- Integrated and troubleshot robot, conveyor, digital I/O, magnetic gripping, sensing, and used CODESYS Depictor visualization for creating Digital Twin, documenting validated operation, failure cases, and system limitations.

Mechatronics & Software Intern Sep 2024 - Mar 2025
Revo Foods GmbH
Vienna, Austria

- Designed and 3D-printed a custom housing, and engineered a motor test station enabling simultaneous performance testing of stepper and servo motors.
- Built a virtual factory simulation in Factory I/O to test and debug PLC code, validating conveyor belt operations and machine sequencing logic.
- Designed and 3D-printed a custom housing, and engineered a motor test station enabling simultaneous performance testing of stepper and servo motors.
- Researched Siemens Sinumerik and other automation systems, and produced schematic diagrams in Draw.io to support planned system upgrades.

Operations Engineer Jan 2023 - Sep 2023
Pro-Spec Performance Parts
Bengaluru, India

- Integrated Amazon FBA (Fulfillment by Amazon) and Amazon MFN (Merchant Fulfilled Network) sales channels with Zoho Books and Zoho Inventory, automating inventory synchronization, order processing, stock visibility.
- Used artificial intelligence-driven demand forecasting, Advanced Excel, and Material Requirements Planning to optimize production planning, stock replenishment, and inventory levels.

OTHER EMPLOYMENT

- **Application Reviewer** - International Office, Hochschule Schmalkalden
  (Jan 2024 - August 2025)
- **Student Assistant** - EDEKA
  (July 2025 - Jan 2026)

HOBBIES

- Studying Philosophy, History
- Cooking new dishes
- Watching Bio-Tech Podcast
- Badminton
- Chess
- Piano

OTHER INFORMATION

- Valid EU Category B Driving Licence
- No Visa/Permit Sponsorship Required.
- Authorized to work in the European Union without visa sponsorship or additional work permits.
- Available to Join Immediately
- Open to travel for work

AWARDS

- Deutschlandstipendium 2024
- IEEE Bangalore Section Best Volunteer
- Zenken Corp. Best Student Volunteer.

PROJECTS

Smart Walker Assistance
Aug 2026 - Present

Carl Zeiss-Stiftung & Hochschule Schmalkalden

- Building a smart-walker safety system with IMU fall detection, GNSS tracking, LTE/SMS alerts, and assisted navigation.
- Developing an Arduino UNO Q with M5Stack Tab5 architecture for real-time sensing, emergency response, and offline-capable operation.
- Developing a Flutter smartphone application prototype to validate fall detection, live location, SOS messaging, and user safety workflows.

PLC-Controlled Pneumatic Automation
Jan 2026 - Feb 2026

Hochschule Schmalkalden

- Designed and implemented sequential PLC control for automated workpiece handling, clamping, positioning, and multi-stage drilling using Siemens S7 PLCs and Ladder Logic (KOP).
- Developed sensor-driven actuator sequencing using Set/Reset logic, digital I/O mapping, intermediate positions, timers, and automatic cycle repetition; later exercises required deriving the control sequence and signal-flow logic from the machine requirements rather than simply reproducing a given program.
- Implemented a more advanced machining cycle with two drilling axes, multi-position workpiece transport, a 1-second inter-operation delay, automatic restart based on part detection, and production counting, then tested the control program on the physical automation setup.

Vision-Guided Delta Robot System
Nov 2025 - Dec 2025

Hochschule Schmalkalden

Video Link - Delta Robot

- Integrated machine vision with a B&R X20 PLC for real-time X–Y object localization and robot target generation.
- Programmed Delta robot motion and pneumatic vacuum gripping for automated part handling.
- Implemented PLC–vision communication and validated the complete pick-and-place cycle on a physical automation cell.

URScript-Based Robotic Assembly
May 2025 - Jul 2025

Hochschule Schmalkalden

Video Link - UR Robot

- Programmed a Universal Robots collaborative manipulator in URScript for automated pick, transfer, alignment, and insertion of components.
- Implemented coordinate-based robot positioning using predefined Cartesian poses, TCP configuration, and controlled approach/retraction motions.
- Programmed joint and linear motion sequences together with gripper control for repeatable component handling and insertion.